MIL-W-22759/11F
10 September 1976
SUPERSEDING
MIL-W-22759/11E
23 April 1976
See also "Supersession Data" on Page 3

MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, EXTRUDED TFE, SILVER-COATED COPPER CONDUCTOR, 600-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.

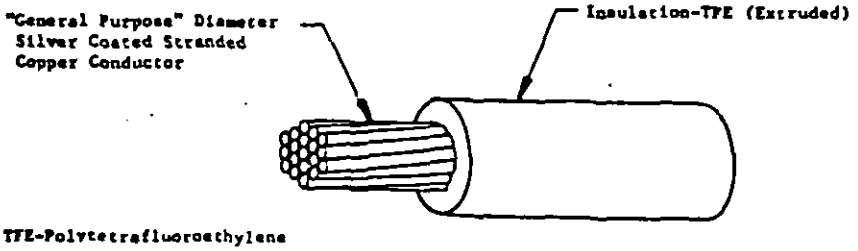


TABLE I. CONSTRUCTION DETAILS

Part No. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M22759/11-28-*	28	7 X 36	.014	.015	63.8	.033 ±.002	1.36
M22759/11-26-*	26	19 X 38	.018	.020	38.4	.038 ±.002	1.90
M22759/11-24-*	24	19 X 36	.023	.025	24.3	.043 ±.002	2.58
M22759/11-22-*	22	19 X 34	.029	.032	15.1	.049 ±.002	3.72
M22759/11-20-*	20	19 X 32	.037	.040	9.19	.058 ±.002	5.43
M22759/11-18-*	18	19 X 30	.046	.050	5.79	.068 ±.002	8.14
M22759/11-16-*	16	19 X 29	.052	.057	4.52	.075 ±.002	10.0
M22759/11-14-*	14	19 X 27	.065	.072	2.88	.090 ±.002	15.1
M22759/11-12-*	12	19 X 25	.082	.090	1.81	.111 ±.003	24.1
M22759/11-10-*	10	37 X 26	.106	.112	1.19	.139 ±.004	37.8
M22759/11-8-*	8	133 X 29	.158	.169	.658	.202 ±.004	65.5

1/ PART NO.: The asterisks in the part number column, Tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/11-20-9; white with orange stripe - M22759/11-20-93.

(F) denotes changes

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:46Z
Check the source to verify that this is the current version before use.

TABLE II. PERFORMANCE DETAILS

Part No.	Abrasion resistance (Procedure II)				Bend testing.			
					Mandrel diameter (inches) (±3%)		Test load (lbs) (±3%)	
	Resistance (inches of tape) (min) (initial condition)	Weight support bracket	Weight (lbs)	Tension load (lbs)	Life cycle (oven & bend tests) 1/	Cold bend test	Life cycle (oven & bend tests) 1/	Cold bend test
M22759/11-28-*	12	A	.50	1.0	.125	.250	.50	
M22759/11-26-*	18	A	.50	1.0	.125	.250	.50	
M22759/11-24-*	18	A	.50	1.0	.125	.250	.50	
M22759/11-22-*	24	A	.50	1.0	.250	.375	.75	
M22759/11-20-*	24	A	.50	1.0	.250	.375	.75	
M22759/11-18-*	20	B	.75	1.0	.250	.375	1.00	
M22759/11-16-*	20	B	.75	2.0	.375	.500	1.00	
M22759/11-14-*	15	B	1.00	2.0	.500	.750	2.00	
M22759/11-12-*	15	B	1.00	2.0	.500	.750	2.00	
M22759/11-10-*	15	B	1.00	2.0	.750	1.00	3.00	
M22759/11-8-*	15	B	1.00	2.0	.750	1.00	3.00	

1/ Also for bend tests after immersion.

ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
SPARK TEST OF PRIMARY INSULATION: Not required
(F) IMPULSE DIELECTRIC TEST: 100% test; 8.0 kilovolts (peak) for sizes 18 and smaller, 6.5 kilovolts (peak) for sizes 16 and larger
INSULATION RESISTANCE: 50,000 megohms for 1000 ft (min)
WRAP TEST: "Wrap back" test required; no cracking
Oven temperature, 313 ±2°C (595.4 ±3.6°F)
BLOCKING: 260 ±2°C (500 ±3.6°F)
SHRINKAGE: 0.03 inch max at 290 ±2°C (554 ±3.6°F)
WICKING: No requirement
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
THERMAL SHOCK: Oven temperature, 200 ±2°C (392 ±3.6°F)
Max change in measurement, 0.06 inch
FLAMMABILITY: Post-flame dielectric test not required
LIFE CYCLE: Oven temperature 275 ±2°C (527 ± 3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
DIELECTRIC TEST AFTER IMMERSION: 2200 volts (rms), 60Hz
ABRASION RESISTANCE AFTER IMMERSION: No requirement
HUMIDITY RESISTANCE: No requirement
SURFACE RESISTANCE: 500 megohm-inches (min), initial and final readings
SMOKE: 290°C (554°F)
ACID RESISTANCE: Required
Dielectric test, 2200 volts (rms), 60Hz
COLOR: In accordance with MIL-STD-104, Class 1; white preferred

COLOR STRIPING OR BANDING DURABILITY: 250 cycles (500 strokes) (min), 500 grams weight
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
WIRE LENGTH REQUIREMENTS: Schedule A
SUPERSESSION DATA: The wire of this specification sheet, by part number, replaces and supersedes the wire of MS21985(AS) (canceled) in accordance with Table III. MS21985(AS) included wire sizes 18 through 16 only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:46Z
Check the source to verify that this is the current version before use.

TABLE III. SUPERSESSION BY PART NUMBER

Part number MS21985(AS)	Part number MIL-W-22759/11
MS21985-28-*	M22759/11-28-*
MS21985-26-*	M22759/11-26-*
MS21985-24-*	M22759/11-24-*
MS21985-22-*	M22759/11-22-*
MS21985-20-*	M22759/11-20-*
MS21985-18-*	M22759/11-18-*
MS21985-16-*	M22759/11-16-*

Custodians:
Navy - AS
Army - EL
Air Force - 99

Review activities:
Navy - EC, OS
Army - ME, MI, MU
Air Force -
DSA - IS
NSA

Preparing activity:
Navy - AS
(Project No. 6145-0701-5)

User activities:
Navy - MC
Army - AT, AV
Air Force - 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:46Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications & Standards
Department (ESSD) Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:46Z
Check the source to verify that this is the current version before use.